

Basic Laws of Boolean Algebra

$A \mid\mid (B \mid\mid C) = (A \mid\mid B) \mid\mid C$ $A \&\& (B \&\& C) = (A \&\& B) \&\& C$	Associative law for or Associative law for and	$A+(B+C) = (A+B)+C$ $A(BC) = (AB)C$
$A \&\& (B \mid\mid C) = (A \&\& B) \mid\mid (A \&\& C)$ $A \mid\mid (B \&\& C) = (A \mid\mid B) \&\& (A \mid\mid C)$	Distributive law for and over or Distributive law for or over and	$A(B+C) = AB+AC$ $A+BC = (A+B)(A+C)$
$A \&\& (A \mid\mid B) = A$ $A \mid\mid (A \&\& B) = A$	Law of Absorption Law of Absorption	$A(A+B)=A$ $A+AB=A$
$!(A \mid\mid B) = !A \&\& !B$ $!(A \&\& B) = !A \mid\mid !B$	DeMorgan's law DeMorgan's law	$\overline{A+B} = \overline{A} * \overline{B}$ $\overline{A*B} = \overline{A} + \overline{B}$
$!(!A) = A$	Double negative law	$\overline{\overline{A}} = A$
$A \mid\mid \sim A = \text{true}$ $A \&\& \sim A = \text{false}$	Law of Complement for or Law of Complement for and	$A + \overline{A} = 1$ $A * \overline{A} = 0$
$A \mid\mid A = A$ $A \&\& A = A$	Idempotence of or Idempotence of and	$A+A=A$ $AA=A$
$A \mid\mid \text{false} = A$ $A \&\& \text{true} = A$	Identity Law of or Identity law of and	$A+0=A$ $A*1=A$
$A \mid\mid \text{true} = \text{true}$ $A \&\& \text{false} = \text{false}$	Law of Union Law of Intersection	$A+1=1$ $A*0=0$
$a^{\wedge}b = (a \mid\mid b) \&\& !(a \&\& b)$ $!(a^{\wedge}b) = (!a \mid\mid b) \&\& (a \mid\mid !b)$	exclusive or (xor function) NOT xor function	a or b is true, but not both both true or both false

Truth Tables

$c = a \&\& b$

a	b	c
0	0	0
0	1	0
1	0	0
1	1	1

$c = a \mid\mid b$

a	b	c
0	0	0
0	1	1
1	0	1
1	1	1

$c = a^{\wedge}b$

a	b	c
0	0	0
0	1	1
1	0	1
1	1	0

$c = !a$

a	b	c
0	0	1
0	1	1
1	0	0
1	1	0

$c = \text{true}$

a	b	c
0	0	1
0	1	1
1	0	1
1	1	1

$c = !(a \&\& b)$

a	b	c
0	0	1
0	1	1
1	0	1
1	1	0

$c = !(a \mid\mid b)$

a	b	c
0	0	1
0	1	0
1	0	0
1	1	0

$c = !(a^{\wedge}b)$

a	b	c
0	0	1
0	1	0
1	0	0
1	1	1

$c = !b$

a	b	c
0	0	1
0	1	0
1	0	1
1	1	0

$c = \text{false}$

a	b	c
0	0	0
0	1	0
1	0	0
1	1	0

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$A \&\& (B \mid\mid C) = (A \&\& B) \mid\mid (A \&\& C)$ $A \mid\mid (B \&\& C) = (A \mid\mid B) \&\& (A \mid\mid C)$	Distributive law for and over or Distributive law for or over and	$A(B+C) = AB+AC$ $A+BC = (A+B)(A+C)$
$A \&\& (A \mid\mid B) = A$ $A \mid\mid (A \&\& B) = A$	Law of Absorption Law of Absorption	$A(A+B)=A$ $A+AB=A$
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$a^{\wedge}b = (a \mid\mid b) \&\& !(a \&\& b)$ $!(a^{\wedge}b) = (!a \mid\mid !b) \&\& (a \mid\mid !b)$	exclusive or (xor function) NOT xor function	a or b is true, but not both both true or both false

1.

- $(A * B) + (A * \sim B) = A$
- $(A + B) * (A + \sim B) = A$

2.

- $A + (\sim A * B) = A + B$
- $A * (\sim A + B) = A * B$

$$A + (\bar{A} \cdot B) = A + B$$

$$(A + B) \cdot (A + \bar{B}) = A$$

$$(A + B) \cdot (A + C) = A + (B \cdot C)$$

$$A + B + (A \cdot \bar{B}) = A + B$$

$$(A \cdot B) + (B \cdot C) + (\bar{B} \cdot C) = (A \cdot B) + C$$

$$(A \cdot B) + (A \cdot C) + (\bar{B} \cdot C) = (A \cdot B) + (\bar{B} \cdot C)$$